

- i. Published PEOs must be clear, concise, assessable and realistic within the context of the available resources.
- ii. PEOs must be consistent with the vision and mission of the institution or the department offering the program.
- iii. Curriculum and teaching-learning processes must support the attainment of PEOs. Justifications must be provided for how these contribute to the attainment of the PEOs.
- iv. A process must be developed to assess the level of attainment of each PEO to evaluate the academic program's effectiveness. Adequate evidence and documentation on the assessment of PEO attainment must be provided. The assessment tools should be indicated, and the way in which these tools are used should be explained.
- v. PEO assessment must lead to the periodic review of PEOs.

Program Outcomes and Assessment 4.8

Program Outcomes (POs) or graduate attributes are narrower statements that describe what students are expected to know and be able to do by the time of graduation. These statements relate to the knowledge, skills and attitudes acquired by students while progressing through the program. The program must demonstrate that by the time of graduation, students have achieved an acceptable minimum level of certain knowledge, skills and behavioral traits. The BAETE specifically requires that students acquire the following graduate attributes:

- a) Apply knowledge of mathematics, natural science, engineering fundamentals and an engineering specialization as specified in K1 to K4 respectively to the solution of complex engineering problems.
- b) Identify, formulate, research literature and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences. (K1 to K4)
- c) Design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations. (K5)
- d) Conduct investigations of complex problems using research-based knowledge (K8) and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.
- e) Create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex engineering problems, with an understanding of the limitations. (K6)

- f) Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex engineering problems. (K7)
- g) Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex engineering problems in societal and environmental contexts. (K7)
- h) Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice. (K7)
- i) Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.
- j) Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- k) Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- l) Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

In addition to incorporating the above-listed POs (graduate attributes), the educational institution may include additional outcomes in its learning programs. An engineering program that aims to attain the abovementioned POs must ensure that its curriculum encompasses all the attributes of the Knowledge Profile (K1 – K8) as presented in Table 4.1 and as included in the PO statements. The ranges of Complex Problem Solving (P1 – P7) and Complex Engineering Activities (A1 – A5) are given in Tables 4.2 and 4.3, respectively.

Table 4.1: Knowledge Profile

Attribute	
K1	A systematic, theory-based understanding of the natural sciences applicable to the discipline
K2	Conceptually based mathematics, numerical analysis, statistics and the formal aspects of computer and information science to support analysis and modeling applicable to the discipline
K3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline
K4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline

K5	Knowledge that supports engineering design in a practice area
K6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline
K7	Comprehension of the role of engineering in society and identified issues in engineering practice in the discipline: ethics and the engineer's professional responsibility to public safety; the impacts of engineering activity; economic, social, cultural, environmental and sustainability
K8	Engagement with selected knowledge in the research literature of the discipline

Table 4.2: Range of Complex Engineering Problem Solving

Attribute	Complex Engineering Problems have characteristic P1 and some or all of P2 to P7:
Depth of knowledge required	P1: Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3, K4, K5, K6 or K8 which allows a fundamentals-based, first principles analytical approach
Range of conflicting requirements	P2: Involve wide-ranging or conflicting technical, engineering and other issues
Depth of analysis required	P3: Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models
Familiarity of issues	P4: Involve infrequently encountered issues
Extent of applicable codes	P5: Are outside problems encompassed by standards and codes of practice for professional engineering
Extent of stakeholder involvement and conflicting requirements	P6: Involve diverse groups of stakeholders with widely varying needs
Interdependence	P7: Are high-level problems including many component parts or sub-problems

Table 4.3: Range of Complex Engineering Activities

Attribute	Complex activities means (engineering) activities or projects that have some or all of the following characteristics:
Range of resources	A1: Involve the use of diverse resources (and for this purpose resources include people, money, equipment, materials, information and technologies)
Level of interaction	A2: Require resolution of significant problems arising from interactions between wide-ranging or conflicting technical, engineering or other issues
Innovation	A3: Involve creative use of engineering principles and research-based knowledge in novel ways
Consequences for society and the environment	A4: Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation
Familiarity	A5: Can extend beyond previous experiences by applying principles-based approaches

Requirements and desirable attributes under this criterion are described in terms of the following sub-criteria.

- i. POs specified by the program must be significantly equivalent to the twelve graduate attributes or POs of BAETE.
- ii. POs must contribute to each PEO.
- iii. The process involved in defining and refining the POs must be described. The correlation between the course outcomes (COs) and POs must be demonstrated through the mapping of COs onto POs.
- iv. The way in which each attribute of the Knowledge Profile (K1 – K8) is addressed in the curriculum must be demonstrated through mapping. The program must also demonstrate how the attributes of the Range of Complex Engineering Problems (P1 – P7) and Complex Engineering Activities (A1 – A5) are incorporated in the teaching, learning and assessment.
- v. A course file must be maintained for each course. The course file should include the assessment of outcomes, curriculum, examination questions and answer scripts, other assessment tools and samples of corresponding student works, and a summary of performance and attainment of course outcomes with suggestions or feedback for future development.
- vi. POs must be assessed using direct methods. Direct methods of assessment are accomplished through the direct examination or observation of students' knowledge or skills against measurable performance indicators or rubrics. In addition, indirect methods may also be used for PO assessment. Indirect methods of assessment are based on opinions or self-reports from different stakeholders. The way in which various assessment tools, including examinations and rubrics, contribute to the evaluation of attainment of each PO must be described. The results of the evaluation of PO attainment must be shown.
- vii. It must be demonstrated through evidence from appropriate evaluation that the students attain all the POs by the time of the graduation.

4.9 Continuous Quality Improvement

The program must have a continuous quality improvement mechanism. Requirements and desirable attributes under this criterion are described in terms of the following sub-criteria.

- i. The program must demonstrate an established system for periodically compiling the level of attainment of PEOs, including a mechanism for tracking and obtaining feedback from graduates and their employers